Branch: MCA (Data Science)	**Semester:** 2nd
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Subject Name: Technical Training - I Lab	**Subject Code**: 25CAP-652
Section/Group: 25MCD- KAR1	**Date of Performance:** 07-04-2026

Experiment No. : 10

1. **Aim of the Session:**
   To understand and apply transaction control in PostgreSQL using BEGIN, COMMIT, ROLLBACK, and SAVEPOINT to maintain data integrity during database operations.

2. **Software Requirements:**
   - PostgreSQL
   - pgAdmin
   - Windows Operating System

3. **Objectives of the Session:**
   Apply transaction control commands for safe and reliable database updates.

4. **Theory:**

Transaction control in PostgreSQL ensures that a group of database operations is executed safely as a single unit. Using **BEGIN**, a transaction starts; **COMMIT** permanently saves the changes, while **ROLLBACK** undoes them if an error occurs. **SAVEPOINT** allows partial rollback to a specific point within a transaction. These commands help maintain data integrity and consistency, especially during complex or multi-step operations.

5. **Practical / Experiment Steps:**
   Step 0: Table creation and data insertion.

   Code:
```
-- Create table
CREATE TABLE payroll (
    emp_id INT PRIMARY KEY,
    emp_name VARCHAR(50),
    salary DECIMAL(10,2) CHECK (salary > 0)
);

INSERT INTO payroll VALUES
```

```
(1, 'Sanju', 30000),
(2, 'Riya', 40000),
(3, 'Sidhi', 50000);

START TRANSACTION;

UPDATE payroll
SET salary = 1000
WHERE emp_id = 3;

UPDATE payroll
SET salary = 1000
WHERE emp_id = 3;

ROLLBACK;

SELECT * FROM payroll;

START TRANSACTION;

UPDATE payroll
SET salary = salary + 5000
WHERE emp_id = 1;

SAVEPOINT sp1;

UPDATE payroll
SET salary = salary + 7000
WHERE emp_id = 2;

UPDATE payroll
SET salary = -1000
WHERE emp_id = 3;

ROLLBACK TO sp1;

COMMIT;

SELECT * FROM payroll;
```

Output:

	emp_id [PK] integer	emp_name character varying (50)	salary numeric (10,2)
1	1	Sanju	30000.00
2	2	Riya	40000.00
3	3	Sidhi	50000.00

	emp_id [PK] integer	emp_name character varying (50)	salary numeric (10,2)
1	2	Riya	40000.00
2	3	Sidhi	50000.00
3	1	Sanju	35000.00

6. **Learning Outcomes (What I have learned):**

- **BEGIN** starts a transaction block for grouping multiple SQL commands.
- **ROLLBACK** cancels changes made inside the transaction.
- **SAVEPOINT** allows partial rollback without losing all valid updates.
- **CHECK** constraint ensures salary remains greater than zero.
- **PostgreSQL** transaction control helps maintain consistency and data integrity.